

NBA opening-spread walk-forward — model and strategy review

Code, data pipeline and the full research register: github.com/sean-qin-usa/nba-prediction-model

Strategy

Directional value-taking against the opening point spread in NBA regular-season games. A market-blind margin model prices every listed game; its margin is compared with the opening spread, and a single side is taken when the disagreement exceeds a walk-forward selected threshold, at the best number available across books. Every position is one side of one game at -110, entered at the open and held to settlement: no hedge, no offsetting leg, no in-game management, no exit. Flat 1 unit per bet, so nothing compounds. No calendar filter.

Selection is the strategy. The book below is 500 bets out of roughly 3,690 regular-season games in the window, about 14% of the slate. A side is taken only where the model's margin disagrees with the opening spread by more than the walk-forward threshold, so the cover rate on the selected book is not the cover rate on the slate.

Simulation: 500 bets, 3 seasons, 2023-24 → 2025-26. The configuration is selected on seasons 1..k from a pre-declared space, frozen, scored on season k+1, and rolled forward; nothing is refitted on the season being scored. The selection history reaches back to 2012-13, so the three seasons here are scored against a decade of prior data rather than against each other. Each bet is priced at the best of the 8 books held, the maximum available to us, with the outlier-realism haircut reported as a second row.

Headline results

season	books/game at open	bets	P&L (u)	ROI	Sharpe (ann.)
2023-24	7.74	179	+5.53u	+3.09%	0.43
2024-25	1.00	186	+28.45u	+15.30%	2.23
2025-26	1.03	135	+9.63u	+7.14%	0.87
2023-26 pooled, modelled best of 8 books	—	500	+43.62u	+8.72%	1.19
2023-26 pooled, after outlier haircut	—	500	+29.82u	+5.96%	0.81

ROI is P&L per unit staked. Sharpe is annualized by $\sqrt{\text{bets per season}}$. Prices are the best of 8 books, but that panel is measured only in 2023-24, so the pooled figures are modelled for 2024-25 and 2025-26. The haircut row charges for the 8.1% of best-of-N prices that sit more than 1.5 points off the next book.



All 500 bets in date order at the opening spread and -110; flat 1u, so the path is a running sum and nothing is re-selected within it. The gap between the lines is 13.80u, the charge for prices a book would limit, lower or void. The raw path troughs at -9.5u by bet 58, about what a zero-edge walk of 179 bets would produce on its own.

Pooled ROI is +8.72% over 500 bets, with a season-clustered 95% interval of [-6.72%, +24.17%] and an MDE80 of 18.5pp. The interval contains zero and 2024-25 alone supplies 65% of the P&L. The frame is also too short to tune on: measured (D187), a null taking the best of five randomly chosen game subsets buys +2.54 ROI points on average, and every strategy filter tested lands inside that band. Resolving an effect this size needs on the order of 36 seasons; there are 3 on the measured panel.

The model

The model outputs a margin, not a probability. A spread is itself a margin forecast, so the sides test involves no devig or probability conversion.

margin = 0.5·four_factors + 0.5·availability_composition + schedule_layer + tank_term
P(home win) = sigmoid(margin / 7.2)

Two independent estimates of team strength, averaged, plus additive context. The model never sees market odds, enforced structurally rather than by convention. Only bet selection sees the price.

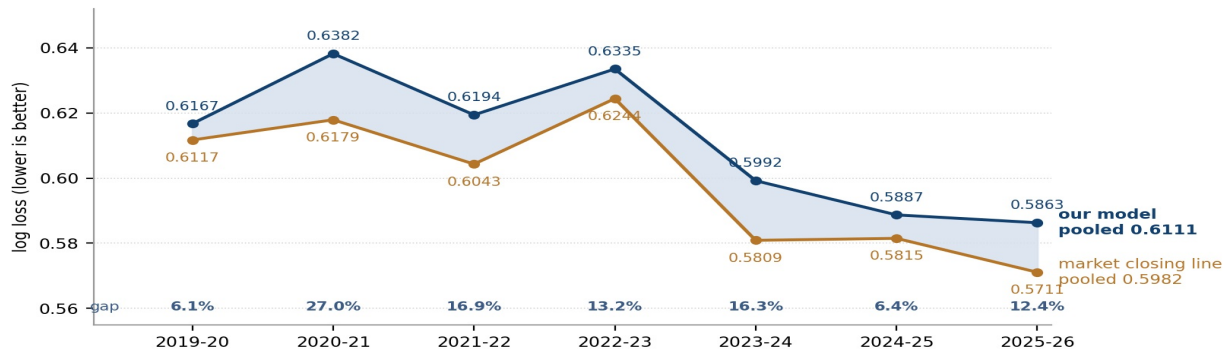
component	what it is	how it is fit
Four factors	opponent-adjusted ratings on shooting, turnovers, rebounding and free-throw rate; a decomposition of points per possession rather than a feature shortlist	one L2 ridge solve per factor (ridge=25), then the four adjusted values mapped to points by a fitted linear map. An 8-factor extension tied, so 4 is kept for parsimony.
Availability composition	Σ over available players of DARKO talent × trailing minutes / 48; the leg that reacts to injuries	requires the daily injury report; this is why the model frame starts 2019-20
Schedule layer	home edge, back-to-backs, dead-team flags	estimated walk-forward, EB shrinkage $n/(n+600)$ toward a prior, team_home_ridge=200 on per-team home deviations. The only component that has survived strict out-of-sample testing on every split tried.
Tank term	late-season effort	exactly zero outside its window

Regularisation, in four forms: L2 ridge on the ratings solve; empirical-Bayes shrinkage toward a prior in the schedule and tank layers; data augmentation via prior-season pseudo-observations; and Bayesian/EB priors in the usage and props fits.

Inputs are four, and all public: nba_api box scores, the daily NBA injury report (PDF ingest), DARKO talent ratings, and historical odds, the last used for selection and scoring only and never by the model. Measured: the entire purchasable stack (professional minutes projections, tracking feeds, premium talent ratings) is worth +0.0012 of log loss combined, not significant. Fitting is restricted to the 002 prefix, 35,546 regular-season games.

Player-level and matchup features were built and gated rather than skipped. The corpus holds a who-guarded-whom matchups cache, defended-FG% by shot category, zone defensive RAPM, hustle counts, ten Synergy play-types, a 10-dimension player skill posterior and 151,914 lineup stints. Tested against the shipped stack, the defensive-RAPM team aggregate returns +0.0011 with an interval crossing zero and the defender-aware variant +0.0037 not significant. Composition built from RAPM sums scores 0.608 against the shipped DARKO version. Physical head-to-head attributes are absent entirely: matchup geometry needs commercial tracking data the project does not hold. The one player-level term that survived is the scalar availability composition above.

Accuracy. Normalized gap is $(ll_{us} - ll_{mkt}) / (ln2 - ll_{mkt})$, the share of the closing line's skill-above-a-coinflip the model fails to capture; zero is parity. On the fully-covered frame (2019-20 → 2025-26, K=7, n=8,286) the model sits 13.59% behind the market closing line, and the closing line beats it in every season of the frame.



Log loss by season on the reported frame, model against the de-vigged market closing line. The closing line is below the model in all seven seasons. Raw log loss is not comparable across seasons because the level moves with game noise, so the normalized gap is shown alongside it.

season	model log loss	closing-line log loss	difference	normalized gap
2019-20 (COVID)	0.6167	0.6117	+0.0050	6.1%
2020-21 (COVID)	0.6382	0.6179	+0.0203	27.0%
2021-22	0.6194	0.6043	+0.0151	16.9%
2022-23	0.6335	0.6244	+0.0091	13.2%
2023-24	0.5992	0.5809	+0.0183	16.3%
2024-25	0.5887	0.5815	+0.0072	6.4%
2025-26	0.5863	0.5711	+0.0152	12.4%
pooled (n=8,286)	0.6111	0.5982	+0.0129	13.59%

The two COVID seasons are the extremes in both directions: 2019-20 is the model's best season on this frame at 6.1% and 2020-21 its worst at 27.0%. Both are kept in because they are fully injury-covered, which is the frame's only criterion.

Two further diagnostics, both on all games rather than the selected book. Against the opening spread the model's margin covers 50.65%, above a coin flip and significant, and below the 52.38% a –110 book charges, also significant. We disagree with the opening line by 2.455 points on average; if that disagreement were entirely real information we would cover 57.6%, so the genuine content is 0.206 points, 8.4% of what we claim. Closing-line value is positive and significant across 19 seasons, so the disagreement does carry information relative to the market's final price, by a margin smaller than the cost of transacting it.

Considerations

Reporting frames

Two frames, each defined by what data exists rather than by which window scored best. Model accuracy: 2019-20 → 2025-26. The daily injury report the availability leg is built on begins 2018-12-17, so 2019-20 is the first fully covered season and earlier seasons score a different, crippled model. Betting: 2023-24 → 2025-26, the era of the measured multi-book panel. The betting frame is not a profitable-window selection: it includes the flattest of the three seasons, and dropping 2023-24 would raise the estimate to +11.86% while widening the interval to ±57 points.

Execution

A separate experiment, not the book above. Model held fixed and only the transacted price varied, on the registered-rule union at the open (n=981, nineteen-season frame, moneyline). It measures the venue rather than the strategy, and its ROI levels are not comparable with the headline.

execution	cost over fair	union ROI	status
1 retail book	3.00 pts	−0.95%	degenerate reference
2 books	—	+0.39%	measured
5 books	1.10 pts	+1.76%	firm baseline
8 books	0.67 pts	+2.39%	max books, the reported tier
exchange, 2% commission	0.41 pts	+2.69%	arithmetic, no exchange data held

The ladder flattens fast: 36% of the time two books post the same number, so extra books duplicate rather than add.

Research record

The register is a record of what was tried and mostly rejected. It is not evidence that the model is good; the accuracy and interval figures above speak to that, and neither is flattering. docs/DECISIONS.md is append-only and runs to D188, with each experiment pre-registered with a SHA-256 hash written before any endpoint is scored, gated out of sample, and recorded whether it worked or not. Feature search on the sides model has since been stopped: nineteen seasons, an exhaustion audit, a 49-feature battery and a possession-level rebuild produced nothing that survived out of sample, which is the reason for stopping rather than a claim that the space is exhausted.

Two results set the bar everything else is read against.

- **Manufacturing capacity.** Tuning to a single season yields positive in-sample ROI on 19 of 19 seasons, mean +15.79% in sample against −1.13% out, a decay of 16.92 ROI points. The identical procedure run on pure noise manufactures +17.46. Net of noise this project's capacity is −0.55 ($p = 0.685$). Every development-versus-out-of-sample gap in the register is smaller than what a modest grid search produces from nothing.
- **Beating a permutation null is necessary, not sufficient.** Three selectors each beat their own null ($p \leq 0.048$ after multiple-comparison correction) and all three lost to the incumbent, one showing +6.29 net-of-null on a paired estimate of −0.43. Net-of-null figures are reported against the incumbent as well as the null.

The register carries errors found and corrected alongside results: an availability leak that made the published expectation too good by 3.8 points of normalized gap, a switch named after a hypothesis its code did not implement, a coefficient significant only because of the 2020 bubble, intervals too narrow before clustering, a team-name join that silently dropped 2,514 rows, and two reporting frames corrected in public. 153 tests cover leakage and reproducibility.

Methodology

- **Walk-forward, no lookahead.** Configuration chosen on seasons 1..k from a pre-declared space, scored on k+1 only.
- **Settlement-only fills.** Priced at the recorded opening spread, held to settlement at −110. No queue model, no partial fill, no re-pricing.
- **Regular season only.** Filtered on the 002 prefix. NBA Cup group-stage games are in because they count in the standings; a pre-registered difference-in-differences found no detectable effect (MDE 2.21 points).
- **Season-clustered inference,** checked across rolling-origin, leave-one-season-out, block-bootstrap and legacy splits, corrected for multiple comparisons across the running family.

Caveats

- Simulated. Research is complete, the model is not in production, no capital has been deployed.
- **Best-of-8 pricing is observed in one season and inferred in two.** A measured multi-book panel exists for 2023-24 only, at 7.74 books per game; 2024-25 observes 1.00 and 2025-26 observes 1.03, so their price comes from a shopping law applied to one recorded book. Those two seasons carry 87% of the window's P&L.
- **Best-of-N transacts at whichever book is furthest offside.** 8.1% of best prices sit more than 1.5 points off the next book, precisely the prices that get limited, lowered or voided. That is what the haircut row charges for. The report otherwise assumes no book limits, voids or re-prices us.
- **No market impact.** The CLV here is a price-taker's number at zero size; real stake into a soft opening line is part of the flow that closes the gap.
- **The model is era-specific by construction.** The walk-forward loop re-selects the betting configuration honestly, but the model architecture was chosen on a 2021-26 corpus and handed to every step as fixed; ablating the era-specific context terms flips the sign of the betting result. Zero shipped components can be dated to a gate using only pre-2021 data. The first season on which the structure is genuinely out of sample is 2026-27.
- **CLV is a monitor, not an objective.** A divergence selector bought more CLV and less ROI; a CLV-targeted selector bought no extra CLV and the most ROI. Green CLV is evidence the prices are good, not that the strategy is profitable.
- Max drawdown, P&L per day and win days require the daily book and are omitted. Exchange fees, state taxes and account limits are not modelled.

Next steps

- **Capture at least two books at the open, from opening night.** Best-of-two lifts CLV by about 49%, and taking the worse book erases nearly all of it. The logger has never run in-season, so if it is not up on opening night the season's open-price CLV record is lost and cannot be reconstructed. Free, and the highest-value operational change available.
- **Pressure-test the walk-forward selection.** The only candidate that clears the vig and beats its noise control. Open questions: whether recency-weighted selection beats all-history, and whether the effect survives a smaller search space. It may only ever be settled live.
- **CLV as the live yardstick, not as the objective.** Significant across 19 seasons and positive in 17, in units that need no devig convention.
- **Exchange access, if it becomes available.** Cost over fair falls from about 3.00 points to 0.41, the largest single lever measured here and an access problem rather than a modelling one.
- **Props before sides.** Both shipped improvements of the last cycle came from the props engine, and soft books are softer on props.

Not next: further feature search on the sides model. The question the project turns on is whether live closing-line value against opening prices, with two books captured, behaves as the backtest says.